

Linear Algebra II (2026 Fall)

Lecture 4.

Studying Contents :

Chapter 5. Diagonalization

- Section 5.2 : Diagonalizability

Section 5.2 Diagonalizability

Recall : A matrix $A \in M_{n \times n}(\mathbb{F})$ is said to be **diagonalizable** if

there exists an ordered basis $\mathcal{B} = \{v_1, \dots, v_n\}$ of $\mathbb{F}^n$ and $\lambda_1, \dots, \lambda_n \in \mathbb{F}$
such that $Av_j = \lambda_j v_j$.

Want to find :

- (i) a simple test to determine whether $A \in M_{n \times n}(\mathbb{F})$ can be diagonalized;
- and
- (ii) a method for actually finding a basis of “eigenvectors”.

Theorem 5.5 : Let $T : V \rightarrow V$ be a linear transformation of an n -dimensional vector space V .

Let $\lambda_1, \dots, \lambda_k \in \mathbb{F}$ be distinct eigenvalues of T and

let $v_1, \dots, v_k \in V$ be eigenvectors of T such that $Tv_j = \lambda_j v_j$ ($1 \leq j \leq k$).

Then $\{v_1, \dots, v_k\}$ is linearly independent.

Proof : (Induction on $k \geq 1$)

$k = 1$: Obvious.

$k \geq 2$: We may assume that $\{v_1, \dots, v_{k-1}\}$ is linearly independent by induction hypothesis.

Suppose that $a_1 v_1 + \dots + a_k v_k = O_V$ for some $a_1, \dots, a_k \in \mathbb{F}$.

Then $T(a_1 v_1 + \dots + a_k v_k) = a_1 \lambda_1 v_1 + \dots + a_k \lambda_k v_k = O_V$.

If $a_k = 0$, then $a_1 = \dots = a_{k-1} = 0$.

Suppose that $a_k \neq 0$, and consider
$$\begin{cases} a_1 \lambda_k v_1 + a_2 \lambda_k v_2 + \dots + a_k \lambda_k v_k = O_V \\ a_1 \lambda_1 v_1 + a_2 \lambda_2 v_2 + \dots + a_k \lambda_k v_k = O_V \end{cases}$$

Then $a_1 (\lambda_k - \lambda_1) v_1 + a_2 (\lambda_k - \lambda_2) v_2 + \dots + a_{k-1} (\lambda_k - \lambda_{k-1}) v_{k-1} = O_V$.

Then $a_1 (\lambda_k - \lambda_1) = a_2 (\lambda_k - \lambda_2) = \dots = a_{k-1} (\lambda_k - \lambda_{k-1}) = 0$.

Then $a_1 = \dots = a_{k-1} = 0$ and hence $a_k = 0$. ■

Theorem 5.5 : Let $T : V \rightarrow V$ be a linear transformation of an n -dimensional vector space V .

Let $\lambda_1, \dots, \lambda_k \in \mathbb{F}$ be distinct eigenvalues of T and

let $v_1, \dots, v_k \in V$ be eigenvectors of T such that $Tv_j = \lambda_j v_j$ ($1 \leq j \leq k$).

Then $\{v_1, \dots, v_k\}$ is linearly independent.

Corollary : Let $T : V \rightarrow V$ be a linear transformation of an n -dimensional vector space V .

If T has n distinct eigenvalues, then T is diagonalizable.

Proof : Let $\lambda_1, \dots, \lambda_n \in \mathbb{F}$ be n distinct eigenvalues of T .

Let $v_1, \dots, v_n \in V$ be eigenvectors of T such that

$$Tv_j = \lambda_j v_j \quad (1 \leq j \leq n).$$

Then $\{v_1, \dots, v_n\}$ is linearly independent by [Theorem 5.5](#).

Therefore $\{v_1, \dots, v_n\}$ is a basis for V . ■

Example 1. : Let

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \in M_{2 \times 2}(\mathbb{R}).$$

Then the characteristic polynomial of A (and hence of L_A) is

$$\det(A - tI_2) = \det \begin{pmatrix} 1-t & 1 \\ 1 & 1-t \end{pmatrix} = t(t-2).$$

Thus L_A has two distinct eigenvalues, 0 and 2, and so A (and hence L_A) is diagonalizable.

Example : Let

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 12 & 1 & 1 \end{pmatrix} \in M_{3 \times 3}(\mathbb{R}).$$

Then the characteristic polynomial of A (and hence of L_A) is

$$\det(A - tI_3) = \det \begin{pmatrix} 1-t & 2 & 3 \\ 2 & 3-t & 1 \\ 12 & 1 & 1-t \end{pmatrix} = -(t^3 - 5t^2 + 34t - 80) = -(t-8)(t-2)(t+5).$$

Thus L_A has three distinct eigenvalues, -5 , 2 and 8 , and so A (and hence L_A) is diagonalizable.

Remark : Let $T : V \rightarrow V$ be a linear transformation of an n -dimensional vector space V .

Suppose that T is diagonalizable. Then does T have n distinct eigenvalues?

Answer : No.

For example, if $A = I_n$ then it is diagonalizable

$$\text{but } \det(A - tI_n) = (1 - t)^n.$$

Definition : We say that a polynomial $f(t) \in P(F)$ **splits over** F if

$$f(t) = c(t-a_1)(t-a_2) \cdots (t-a_n) \text{ for some } c \in F \text{ and } a_1, \dots, a_n \in F \text{ (not necessarily distinct).}$$

Examples and Remark :

(1) $f(t) = t^2 - 1 = (t-1)(t+1)$ splits over $\mathbb{R}$.

(2) $f(t) = t^3 - 2t^2 + t - 2 = (t^2 + 1)(t-2)$ does not split over $\mathbb{R}$,

but it splits over $\mathbb{C}$ since $f(t) = (t^2 + 1)(t-2) = (t+i)(t-i)(t-2)$.

Thus the property that a polynomial $f(t) \in P(F)$ depends heavily on the field F .

(3) Let V be a finite dimensional vector space over a field F , and

let $T: V \rightarrow V$ be a linear transformation.

Also, let $f(t) := \det(T - tId_V)$ be the characteristic polynomial of T .

Then we say that $f(t)$ **splits** if it splits over F .

Theorem 5.6. : Let $T : V \rightarrow V$ be a linear transformation of an n -dimensional vector space V .

If T is diagonalizable, then its characteristic polynomial splits.

Proof : Let β be an ordered basis for V such that $D := [T]_{\beta}$ is a diagonal matrix.

$$D = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$$

If $\lambda_1, \dots, \lambda_n$ are the diagonal entries of D , then

$D - tI_n$ is the diagonal matrix whose diagonal entries are $\lambda_1 - t, \dots, \lambda_n - t$.

Therefore the characteristic polynomial of T is equal to

$$(\lambda_1 - t) \cdots (\lambda_n - t) = (-1)^n (t - \lambda_1) \cdots (t - \lambda_n).$$



Summary : Let V be an n -dimensional vector space over a field F , and consider the map

$$\mathcal{L}(V) \rightarrow P_n(F), \quad T: V \rightarrow V \mapsto \det(T - tId_V) =: f(t).$$

- (1) If $f(t)$ has n -distinct zeros in F , then T is diagonalizable and $f(t)$ splits (over F).
- (2) If T is diagonalizable, then $f(t)$ splits (over F).
- (3) It may happen that $f(t)$ splits over F but T fails to be diagonalizable.
- (4) It may happen that T is diagonalizable but $f(t)$ does not have n -distinct zeros in F .

$f(t)$ has n -distinct zeros in F .

YES $\Downarrow \Uparrow$ NO (There exists a counter-example.)

T is diagonalizable.

YES $\Downarrow \Uparrow$ NO (There exists a counter-example. – see [Example 3.](#))

$f(t)$ splits over F .

Definition : Let $T : V \rightarrow V$ be a linear transformation of an n -dimensional vector space V and let λ be an eigenvalue of T .

(1) The **(algebraic) multiplicity of** λ is defined to be

the largest integer k for which $(t - \lambda)^k$ is a factor of the characteristic polynomial of T .

(2) The set $E_\lambda := \{x \in V \mid T(x) = \lambda x\}$ is called

the eigenspace of T corresponding to the eigenvalue λ .

(When A is an $n \times n$ square matrix, we define its eigenspaces as those of L_A .)

Remark : (1) E_λ is a subspace of V since $E_\lambda = N(T - \lambda I_V)$.

(2) E_λ consists of O_V and all eigenvectors corresponding to the eigenvalue λ .

(3) The dimension of E_λ is equal to

the maximum number of linearly independent eigenvectors of T corresponding to the eigenvalue λ .

Theorem 5.7. : Let $T : V \rightarrow V$ be a linear transformation of an n -dimensional vector space V and let λ be an eigenvalue of T having multiplicity m . Then

$$1 \leq \dim E_\lambda \leq m.$$

Proof : Choose an ordered basis $\{v_1, v_2, \dots, v_p\}$ for E_λ .

Then extend it to a basis $\beta = \{v_1, \dots, v_p, v_{p+1}, \dots, v_n\}$ for V .

Then the matrix $A := [T]_\beta = [[T(v_1)]_\beta [T(v_2)]_\beta \cdots [T(v_n)]_\beta]$ is of the following form :

$$A = \begin{pmatrix} \lambda I_p & B \\ O & C \end{pmatrix}$$

Thus the characteristic polynomial $f(t)$ of T is

$$\begin{aligned} f(t) &= \det(A - tI_n) = \det \begin{pmatrix} (\lambda - t)I_p & B \\ O & C - tI_{n-p} \end{pmatrix} \\ &= \det((\lambda - t)I_p) \det(C - tI_{n-p}) \\ &= (\lambda - t)^p g(t), \end{aligned}$$

where $g(t)$ is a polynomial.

This shows that $(t - \lambda)^p$ is a factor of $f(t)$ and hence the multiplicity of λ is at least p .

Since we put $p = \dim E_\lambda$, it follows that $\dim E_\lambda \leq m$. ■

Example 2. : For

$$A = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 3 & 4 \\ 0 & 0 & 4 \end{pmatrix},$$

the characteristic polynomial is equal to $f(t) = -(t-3)^2(t-4)$.

Hence $\lambda = 3$ is an eigenvalue of A with multiplicity 2, and

$\lambda = 4$ is an eigenvalue of A with multiplicity 1.

Example 3 : Let $T : P_2(\mathbb{R}) \rightarrow P_2(\mathbb{R})$, $f(t) \mapsto f'(t)$ and let $\beta = \{1, x, x^2\}$. Then

$$[T]_{\beta} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

Thus the characteristic polynomial of T is $\det([T]_{\beta} - tI) = -t^3$.

In particular, $\lambda = 0$ is the only eigenvalue of T (with multiplicity 3).

Also, $E_{\lambda} = E_0 = N(T - \lambda I) = N(T)$ is equal to the space of constant polynomials in $P_2(\mathbb{R})$.

Consequently, there is no basis for $P_2(\mathbb{R})$ consisting of eigenvectors of T .

Therefore T is not diagonalizable.

Example 4. : Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear transformation defined by

$$T \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} 4a_1 & + & a_3 \\ 2a_1 + 3a_2 + 2a_3 \\ a_1 & + & 4a_3 \end{pmatrix}.$$

In this example, we want to find the eigenspaces corresponding to each eigenvalue of T .

(1) Let $\beta = \{e_1, e_2, e_3\}$ be the standard basis for $\mathbb{R}^3$. Then

$$[T]_{\beta} = \begin{pmatrix} 4 & 0 & 1 \\ 2 & 3 & 2 \\ 1 & 0 & 4 \end{pmatrix},$$

and hence the characteristic polynomial of T is

$$\det([T]_{\beta} - tI) = \det \begin{pmatrix} 4-t & 0 & 1 \\ 2 & 3-t & 2 \\ 1 & 0 & 4-t \end{pmatrix} = -(t-5)(t-3)^2.$$

So, the eigenvalues of T are $\lambda_1 = 5$ and $\lambda_2 = 3$ with multiplicities 1 and 2, respectively.

Example 4. (Continued) :

(2) Since

$$E_{\lambda_1} = N(T - \lambda_1 I) = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \in \mathbb{R}^3 : \begin{pmatrix} -1 & 0 & 1 \\ 2 & -2 & 2 \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right\},$$

it follows that E_{λ_1} is the solution space of the system of linear equations

$$\begin{aligned} -x_1 &+ x_3 = 0 \\ 2x_1 - 2x_2 + 2x_3 &= 0 \\ x_1 &- x_3 = 0. \end{aligned}$$

One can check easily that

$$E_{\lambda_1} = \left\{ c \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \mid c \in F \right\} \text{ and hence } \dim E_{\lambda_1} = 1 \text{ (= the multiplicity of } \lambda_1 = 5).$$

(3) Since $E_{\lambda_2} = N([T]_{\beta} - \lambda_2 I)$ is the solution space of the system of linear equations

$$\begin{aligned} x_1 + x_3 &= 0 \\ 2x_1 + 2x_3 &= 0 \\ x_1 + x_3 &= 0 \end{aligned}$$

it follows that

$$E_{\lambda_2} = \left\{ c \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + d \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \mid c, d \in F \right\} \text{ and hence } \dim E_{\lambda_2} = 2 \text{ (= the multiplicity of } \lambda_2 = 3).$$

Example 4. (Continued) :

(4) The following set of eigenvectors of T is a basis for $\mathbb{R}^3$.

$$\left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}$$

Consequently, T is diagonalizable.

Remark : In Example 3,

- (i) the characteristic polynomial of T splits, but
- (ii) the dimension of an eigenspace is strictly less than the multiplicity.

In this case, T fails to be diagonalizable.

In Example 4,

- (i) the characteristic polynomial of T splits, and
- (ii) the dimension of each eigenspace is equal to the multiplicity of the corresponding eigenvalue.

In this case, T fails to be diagonalizable.

This is indeed true by the following Theorem 5.8.

Theorem 5.8. : Let $T : V \rightarrow V$ be a linear transformation of an n -dimensional vector space V .

Also let $\lambda_1, \dots, \lambda_k$ be distinct eigenvalues of T .

For each $1 \leq i \leq k$, let $S_i \subset E_{\lambda_i}$ be a finite linearly independent subset.

Then $S = S_1 \cup S_2 \cup \dots \cup S_k$ is a linearly independent subset of V .